

# Share-faithful reconstruction of bilateral trade from raw UN Comtrade — and a pre-registered nowcast for critical materials

Independent public-data research · [comtrade-reconcile](#) · [interactive atlas](#)

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**In plain terms.** I rebuild who really trades 32 critical raw materials from raw UN Comtrade, check it against the official CEPII BACI data, and find that in **17 of 32** the country that *exports* the most isn't the one that *mines* the most — so standard import statistics misread where supply is concentrated. I extend the series back to **2002**, **test** the trends rather than just plotting them, and **pre-register** the forecast. Everything is public and reproduces from the committed code, no API key.

▼ Full abstract

I reconstruct bilateral trade in 32 critical raw materials from raw UN Comtrade using a Gaulier–Zignago-style mirror reconciliation (reliability-weighted inverse-variance averaging of exporter and importer reports), and validate it against the official CEPII BACI dataset on the statistics that matter for supply-concentration analysis: top exporters, trade shares, and Herfindahl concentration. On 2024 the reconstruction recovers the top exporter in **25 of 30** materials with a share mean-absolute-error of **3.5pp** and an HHI correlation of **0.92** (importer side: 22/30, 4.2pp, 0.97). I then (i) document an **origin gap** — in **17 of 32** materials the top *exporter* is not the top *miner*, so import-origin statistics systematically overstate the geographic diversification of supply — and (ii) build a provisional 2025 nowcast and a directional 2026 scenario whose shared assumption (share persistence) I test out-of-sample (**85%** year-over-year top-exporter persistence; a leader's annual share move has P50 3.5pp / P90 8.7pp) and whose accuracy I **pre-register** against the next BACI release. All inputs are public; the raw Comtrade is committed, so all code and fixtures reproduce the reconciliation and validation end-to-end with no API key. Finally I extend the reconciled series back to **2002** and, rather than merely plotting concentration as the literature does, **test** it: **9 of 32** materials show a

statistically significant rising export-concentration trend (Mann–Kendall, FDR-corrected), with structural breaks clustering in **2012–2016**.

## 1. The question

Under customs rules, refining and substantial transformation confer "origin": when a country imports an ore, processes it and re-exports, the traded good takes the processor's nationality and the customs record stops there. Trade statistics therefore measure *where a material was last shipped from*, not *where it came out of the ground*. For critical-material supply-risk analysis — where the policy question is genuine geographic concentration of the upstream — this is a systematic blind spot. This note builds the data to measure it and quantifies how large the gap is.

## 2. Data and vintages

Table 1 — sources and exact vintages. All public.

| LAYER                                      | SOURCE                           | VINTAGE / COVERAGE                                      |
|--------------------------------------------|----------------------------------|---------------------------------------------------------|
| Reconciled bilateral trade (reference)     | CEPII BACI                       | HS17 release <b>V202601</b> (Jan 2026), years 2018–2024 |
| Raw bilateral trade (reconstruction input) | UN Comtrade API                  | annual + monthly, reporter×partner, pulled 2025–26      |
| Mine production shares                     | USGS Mineral Commodity Summaries | latest published (approx.)                              |
| Refining / processing shares               | IEA Critical Minerals Outlook    | latest published (approx.)                              |
| Reserves                                   | USGS Mineral Commodity Summaries | economically recoverable (approx.)                      |
| Gravity covariates (CIF/FOB)               | CEPII <code>dist_ceprii</code>   | distance, contiguity                                    |
| Commodity prices (2026 tilt)               | World Bank Pink Sheet            | monthly, Q1-2026 vs Q1-2025                             |

The 32 materials are mapped to HS6 product codes; three of them — gallium, germanium and hafnium — share a single HS6 code (811292) and so carry *identical* trade columns that cannot be separated. This is flagged wherever those materials appear.

### 3. Reconciliation method

For each product and country pair  $(i, j)$  two reports exist: the exporter's free-on-board value  $x_{ij}$  and the importer's cost-insurance-freight value  $m_{ji}$ . Following Gaulier & Zignago (2010), I (a) strip an estimated CIF/FOB markup from import values, (b) weight each reporter by an estimate of its reliability, and (c) average the two reports as an inverse-variance weighted mean in logs.

#### CIF/FOB markup

The markup is recovered from the within-product mirror ratio. On the 32-material slice a gravity regression of  $\log(m/x)$  on distance and contiguity is not identified ( $R^2 \approx 0.01$ ) — BACI estimates this on the full ~5,000-product universe — so I fall back to a robust per-product median markup, the same quantity in reduced form.

#### Reliability weights (variance components)

Let  $d_{ij} = \log x_{ij} - \log m_{ji}$  be the mirror discrepancy. Modelling it as the sum of a reporter- $i$  and reporter- $j$  error, an OLS regression of  $d^2$  on reporter dummies yields each reporter's error variance  $v_i$  (a variance-components estimator). The reconciled value averages the two reports with weights inversely proportional to those variances:

$$\log t_{ij} = (w_i \cdot \log x_{ij} + w_j \cdot \log \tilde{m}_{ji}) / (w_i + w_j), \quad w_i = 1 / v_i$$

where  $\tilde{m}$  is the import value after removing the CIF/FOB markup. The result is a single reconciled bilateral matrix per product per year — the same object BACI produces, built here from the raw reports.

### 4. Validation against BACI

The reconstruction is validated on the statistics the atlas actually uses — shares, ranks, concentration — not on absolute levels.

Table 2 — reconstruction vs official BACI, both sides of the market.

| YEAR · SIDE     | TOP - 1 EXPORTER/IMPORTER CORRECT | SHARE MAE | HHI CORRELATION |
|-----------------|-----------------------------------|-----------|-----------------|
| 2024 · exporter | 25 / 30                           | 3.5 pp    | 0.92            |
| 2024 · importer | 22 / 30                           | 4.2 pp    | 0.97            |
| 2022 · exporter | 22 / 30                           | 3.9 pp    | 0.89            |

**On levels.** Reconstructed totals run ~1.5–1.8× BACI. This is *not* a reconciliation artefact: current raw Comtrade already sits ~1.8–1.9× above BACI's published snapshot before any reconciliation (Comtrade absorbs continuous revisions; BACI applies additional outlier/quality filtering this note does not replicate), and the reconstruction sits *between* the two raw mirror reports. A settled year (2022) still shows ~1.5×, so the offset is partly persistent filtering, not only vintage. The claim is therefore explicitly **share-faithful**, not level-faithful — which is the relevant property for concentration analysis.

## 5. Nowcast and pre-registration

BACI lags ~18 months. For the years it has not released I extend the series:

- **2025 (provisional).** The full reconciliation applied to the partial 2025 Comtrade that has been filed (~half of reporters by mid-2026), with levels calibrated per material to BACI 2024.
- **2026 (directional).** Only ~Q1 monthly Comtrade exists, so 2025's reconciled structure is carried forward and only *levels* are tilted, per material, by reporter-matched Q1 export momentum blended with the World Bank Pink Sheet price change. Shares are held at 2025: this is a directional scenario, not a bilateral measurement.

Both rest on one assumption — that last year's trade structure predicts this year's. I test it directly on the measured 2018–2024 series, scoring year  $T-1$ 's shares as a nowcast for year  $T$ :

Table 3 — out-of-sample persistence (192 material-years, 2018–2024).

| METRIC                                    | VALUE   |
|-------------------------------------------|---------|
| Top-exporter unchanged year-over-year     | 85%     |
| Mean share MAE across exporters           | 0.37 pp |
| Leader's annual share move — median (P50) | 3.5 pp  |
| Leader's annual share move — P90          | 8.7 pp  |

The P50/P90 are the nowcast's empirical uncertainty band, shown on the interactive year slider. Most-predictable materials: magnets, coking coal, manganese, niobium; least: arsenic, beryllium, hafnium, gallium, germanium, fluorspar (thin or non-reporting-producer markets).

**Pre-registration.** Before BACI 2025 exists, I have committed ( `PREREGISTRATION.md` ), in git history at the relevant commit) to how the frozen `flows_2025.json` will be scored when it releases — `validate.py 2025` , same metrics — with numeric thresholds ( $\geq 22/30$  top-1,  $\leq 5$ pp share MAE,  $\geq 0.88$  HHI correlation; scored on the 30 validatable materials) and a commitment to publish the result pass or fail. The bet is written down first; the test is falsifiable on a fixed date by one command.

## 6. Finding: the origin gap

With reconciled trade and the mine-production layer side by side, define, per material in a given year:

$$\text{origin gap} = (\text{top exporter's share of world trade}) - (\text{that same country's share of world mine output})$$

A large positive gap means a country sells far more than it mines — it processes or trans-ships someone else's ore. The headline result, on 2024: in **17 of 32** materials the top exporter is not the top miner; in **4 of 32** a country mining under 5% of world supply exports more than a quarter of it.

Table 4 — largest origin gaps, 2024 (where exporter ≠ miner).

| MATERIAL                   | TOP EXPORTER  | EXPORTS | IT MINES | GAP | LEAD MINER          |
|----------------------------|---------------|---------|----------|-----|---------------------|
| Beryllium, unwrought       | Kazakhstan    | 89%     | 0%       | +89 | United States (58%) |
| Strontium carbonate        | Germany       | 60%     | 0%       | +60 | Iran (38%)          |
| Lithium carbonate          | Chile         | 75%     | 24%      | +51 | Australia (48%)     |
| Aluminium ores / bauxite   | Guinea        | 72%     | 24%      | +48 | Australia (24%)     |
| Phosphorus                 | Vietnam       | 47%     | 1%       | +46 | China (41%)         |
| Cobalt oxides & hydroxides | Finland       | 29%     | 0%       | +29 | DR Congo (74%)      |
| Tantalum, unwrought        | United States | 22%     | 0%       | +22 | DR Congo (41%)      |
| Nickel, unwrought          | Norway        | 18%     | 0%       | +18 | Indonesia (50%)     |

**The twist.** The intuitive story is "China hides behind refineries." The data only half-supports it: for many materials China is *both* the lead miner and the lead exporter — its position is largely genuine, not an artefact. The materials where exporter and miner diverge are instead fronted by industrial refiners and entrepôts — Finland for Congolese cobalt, Japan for largely Chinese-mined titanium, Germany for Iranian strontium, Norway for Indonesian nickel. The corrective therefore cuts both ways: it deflates apparent dependence on refiner countries, and it reveals that genuinely concentrated upstreams (Congo cobalt, Indonesian nickel) are *more* concentrated than the diversified-looking trade ledger suggests.

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## 7. Trends, breaks and tested concentration (2002–2024)

The reconciled series is extended back to **2002** using CEPII BACI's HS2002 release, spliced to HS17 from 2017. The 32 materials' HS6 codes are stable across that window with two documented exceptions: boron, whose code was merged in HS2012 (recovered by summing its two predecessor codes), and hafnium, pooled inside 811292 before HS2022 (as gallium and germanium remain). This yields a continuous 22-year, share-faithful panel — long enough to *test*, not merely plot, how concentration evolved. Most critical-minerals work stops at plotting Herfindahl trajectories; here each material's export-HHI, China-share and origin-gap series is put through standard trend statistics:

- **Monotonic trend.** The Mann–Kendall rank test (Mann 1945; Kendall 1975) for significance and the Theil–Sen estimator (Sen 1968) for a robust slope, with Benjamini–Hochberg (1995) false-discovery-rate correction across the 32 materials.
- **Structural breaks.** The Pettitt (1979) non-parametric change-point test dates the year a series shifts regime (cf. Bai & Perron 1998 for multiple breaks).
- **Decomposition.** The 2002→2024 change in export-HHI is attributed by country (contribution =  $\text{share}^2_{2024} - \text{share}^2_{2002}$ ), identifying which exporter drove each material's concentration.

The result: **9 of 32** materials show a statistically significant **rising export-concentration trend** (FDR<0.05); their structural breaks **cluster in 2012–2016**, the period in which export-control measures and critical-minerals-security policy intensified. The steepest are beryllium, tungsten, bauxite and rare-earth magnets. The aggregate **origin gap** — averaged across materials — is wide and somewhat higher at the end of the period (~15→~18pp), but the index steps at the **2017 HS-vintage join**, so this is read as trade-side drift, not a clean trend — an HS17-only confirmation is the proper test — now carried out on the **robustness page** (autocorrelation-robust Mann–Kendall plus the HS02-vs-HS17 sub-period check). Because USGS mine shares enter as a current fixed reference, this isolates how the *trade* map drifted from today's mining geography; it is a trade-side, not a geological, trend. I did not find a peer-reviewed study applying this combined Mann–Kendall / Theil–Sen / Pettitt screening to critical-material export-concentration panels; formal trend and change-point inference on these series appears uncommon in the policy literature. The contribution is the testing, not the plotting.

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## 8. Further analytical layers

Four additional lenses are built on the same reconciled trade, each deliberately framed as a *screening indicator*, not a structural claim:

- **Trade-routing centrality.** Treating each material's bilateral flows as a directed network, betweenness and PageRank (Freeman 1977) plus a **node-removal fragility test** surface the brokers and processing hubs that share-concentration misses. Read as routing centrality — betweenness

on a trade network also reflects import size, so a large buyer scores high without "controlling" anything.

- **Governance-weighted criticality.** An EU/SCRREEN-shaped supply-risk proxy (Blengini et al. 2017) weighting concentration by World Bank Worldwide Governance Indicators, plus a Graedel-style supply-risk axis (Graedel et al. 2012). These are proxies on public approximations, not the official EU or Yale scores; they correlate with the project's transparent index at  $\rho \approx 0.79$  — partly mechanically, since they share inputs.
- **Economic complexity.** Balassa (1965) revealed comparative advantage over the 32-material export matrix — specialization *within* critical materials, not economy-wide RCA — with material ubiquity and relatedness (Hidalgo & Hausmann 2009).
- **Origin trace.** A first-order re-attribution of refiner-fronted imports to the dominant mine, reported explicitly as an *upper bound* on single-origin concentration.

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## 9. Limitations and non-claims

- The origin gap is a **measurement gap** between two public datasets (reconciled trade shares vs USGS mine shares), *not* a fully traced physical supply chain. Some gap is legitimate — a refiner genuinely exports a *different product* than the mined ore (cobalt chemicals vs cobalt ore), which sits under a different code.
- Customs data cannot separate a true refiner from a re-export hub; trade-derived country roles are *trade exposure*, not proof of physical processing. Entrepôts are flagged.
- gallium / germanium / hafnium share one HS6 code and cannot be separated in trade.
- Figures are share-faithful, not level-faithful; absolute values are not claimed. Mine and refine layers are approximate USGS/IEA reference shares, undated. 2025 is provisional; 2026 is directional.

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## 10. Reproducibility

Two public repositories, **no API key required to reproduce the pipeline end-to-end**: the interactive atlas ([materials-atlas/critical-materials-atlas](#)) and the reconciliation engine ([materials-atlas/critical-materials-atlas \(reconcile/\)](#), CI green). The raw 2024 Comtrade is committed (gzipped) under `fixtures/raw/`, so the reconciliation regenerates from raw rather than from a pre-computed file. On a fresh clone of the engine: `pip install -r requirements.txt`, then `ATLAS_ROOT=fixtures python reconcile.py 2024` (regenerates the reconciliation), `... validate.py 2024` (Table 2), `python backtest.py` (Table 3), `python findings.py` (Table 4). CI runs the `reconcile → validate` chain on every push. The Comtrade key is needed only for the initial network pull and is read from the environment, never committed.

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